

Prognosis

Cox Regression and Weibull Accelerated Failure Time (AFT)
[Prognostic mpenrol for predicting Alzheimer's disease conversion using functional connectome manifolds, Kim et al. 2024]

[Detecting Mild Cognitive Impairment to Alzheimer's Disease Progression by fMRI using Convolutional Neural Network and Long-Short Term Memory, Ghafjori et al. 2025]
Unreviewed!

Diagnosis

CNNu/3D-CNNs + LSTM
[Functional MRI and ApoE4 genotype for predicting cognitive decline in amyloid positive individuals, Zhu et al. 2022]

LSTMs
[Symmetry-Aware LSTM-Based Effective Connectivity Framework for Identifying MCI Progression and Reversion with Resting-State fMRI, Sun et al. (2025)]
Architecture: LSTM-Layer-LSTM-ConvNet (LSTM-LSTM)
Inputs: rs-fMRI time series processed for effective (directional) connectivity
Task: Classify MCI progression (Progression vs. Stable vs. Reversion). **Classification not Progressed!**
Dataset: ADNI rs-fMRI
Performance: "84.9% (3-way validation), "90.9% (Reversion vs. Progression).
Key Note: Successfully established directional difference between brain regions rather than data correlation.

[Temporal and Spatial Analysis of Alzheimer's Disease Based on an Improved Convolutional Neural Network and a Resting-State fMRI Brain Functional Network, Sun et al. 2022]
Paper uses an LSTM which last layer could be changed to a regression layer.

CNNu/3D-CNNs
[Predicting conversion from MCI to AD by integration of rs-fMRI and clinical information using 3D-CNN, Ghafjori et al. 2022]
Architecture: 3D-CNN (spatial only) + Dense fusion layers.
Inputs: Single volume rs-fMRI + clinical (Demographics).
Task: Binary 5-year MCI-to-AD conversion.
Dataset: ADNI MCI subset (50 subjects).
Performance: "87.6% Accuracy, "93.7% AUC (with clinical data).
Key Note: Presence of the 3D-CNN truly helped the LSTM temporal modeling component.
Did not read the paper yet!

Spatiotemporal GNN
[A novel spatiotemporal graph convolutional network framework for functional brain network analysis and interpretable cognitive disorder diagnosis, Zhang et al. 2024]
Architecture: Spatiotemporal Graph Convolutional Network with Graph-based Time Activation Mapping (STGC-GCN)
Inputs: rs-fMRI functional connectivity (in clinical validation with T1 for registration only (just as model input))
Task: AD/MCI classification and pMCI conversion prediction within 3 years
Dataset: SUCCORE project and ADRG
Note: Uses fMRI for testing, T1 used only for preprocessing/registration
Dynamic Functional Connectivity

Generalizable

Latent Models for Dynamic Temporal Prediction of Conversion
[Reflections on dynamic prediction of Alzheimer's disease: advancements in modeling longitudinal outcomes and time-to-event data, Chen et al. 2025]
Has fMRI data as input

Joint-Embedding Transformer
[Brain-JEPA: A brain dynamics foundation model with Joint-Embedding Predictive Architecture, Dong, Z., et al. (2024)]

[Interpretable LSTM Model Reveals Transiently-Realized Patterns for Early aMCI Prediction, Gao et al., 2022]
Architecture: Time-Aware LSTM (TA-LSTM)
Inputs: rs-fMRI (symmetrically rescaled) based dynamic functional network connectivity - 1000x50x50
Task: Predict progression or recovery from questionable/mild MCI (pMCI) after 3 years
Dataset: MIND-stc data
Note: Uses dynamic functional network connectivity from fMRI; interpretable model with attention mechanisms

[BrainRGin: Brain ROI-aware Graph Isomorphism Networks, Thapaliya et al., 2025]
Architecture: Graph Isomorphism Network (GIN) with clustering-based embedding + attention-based readout
Inputs: rs-fMRI (static functional network connectivity matrices)
Task: Long-range prediction (generalizable to rs-fMRI)
Dataset: AddNeuron Brain Cognitive Development Dataset
Note: Novel architecture combining GIN with Topol-pooling and attention mechanisms; demonstrates potential for AD applications

Graph Generation
[FBNETGEN: Task-aware GNN-based fMRI Analysis via Functional Brain Network Generation, Cui et al. 2022]
Architecture: Task-aware Graph Neural Network with functional brain network generation
Inputs: rs-fMRI only (generates functional brain networks end-to-end)
Task: Clinical predictions from fMRI
Dataset: Multiple fMRI datasets
Note: Forecasts ROI feature extraction, brain network generation, and prediction with GNNs in an end-to-end trainable framework

fMRI
[Cortical graph neural network for AD and MCI diagnosis and transfer learning, Wee et al. 2019]
Architecture: Spatial Graph Convolutional Neural Network Graph-GNN
Inputs: Cortical thickness and geometry (use fMRI-based, use structural data)
Performance: MCI to AD conversion prediction: 75%, MCI to AD: 92%
Dataset: ADNI, L, ADNI, L, ADNI subset
Note: Uses graph structure but with structural rather than functional data - included for architectural reference

Clinical Data
[Predictive Modeling of the Progression of Alzheimer's Disease Using Deep Recurrent Neural Networks, Wang et al., 2018]
Architecture: LSTM-RNN with Time Intervals LSTM with TE
Inputs: Clinical longitudinal data (just use rs-fMRI, but demonstrates RNN architecture for AD progression)
Performance: >90% accuracy for next-visit AD-stage prediction
Dataset: ADNI
Note: Handles irregular time intervals between visits

GNN
[Spatio-temporal dynamic functional brain network for mild cognitive impairment diagnosis, Wen et al. 2025]
Architecture: Novel spatio-temporal GNN with graph-recurrent neural network (GRNN)
Inputs: rs-fMRI dynamic functional connectivity
Performance: >90% accuracy, BA-25 AUC: 94% vs. AUC
Dataset: ADRG
Task: Identification of specific anatomical regions - left Hippocampus, amygdala, and insula - as key nodes in the dynamic network
Note: Captures dynamic changes across time windows; integrates structural fMRI and EEG in extended version but core model uses fMRI

GNNs (TL-MRI + fMRI + Genetics + Clinical Scores)
[MAGNET-AD: Multitask Spatiotemporal GNN for Interpretable Prediction of PACC and Conversion Time in Preclinical Alzheimer, Hossain, S., et al. 2025]

Classification
aMCI vs pMCI
[An Interpretable Brain Network Atlas-Based Hybrid Model for Mild Cognitive Impairment Progression Prediction, Xianglong Guo, Li Ma, Yanyue Huang, Suqin Tang, and Yinghui Li. 2023]
Classifies based on two labels of pMCI and aMCI

[Brain functional connectivity analysis of fMRI-based classification, Alarjani et al. 2025]
Architecture: Multivariate Pattern Analysis (MPA) + Support Learning Machine (SLM) classifier
Inputs: Rest functional connectivity features
Task: AD stage classification (HC vs. MCI vs. AD)
Dataset: CNRS and ADRG cohorts
Performance: "98.0% Multi-class accuracy (ADRG).
Key Note: Non-deep learning approach; computationally efficient and independent Decision Trees (MD)

[Classification of Alzheimer's progression using fMRI data, Noh et al. 2023] + **CNNu**
Used an encoder-decoder U-Net (U-Net) to extract spatial features from rs-fMRI, coupled with an LSTM to capture temporal dynamics. This U-Net-LSTM model demonstrated 92%, 95%, 95%, and 92% accuracy for MCI, pMCI, and AD subjects for fMRI-based AD Staging/Diagnosis, not for conversion prediction!

[Multi-label classification of Alzheimer's disease stages from resting-state fMRI, Alorfi, 2022]
Architecture: Deep learning with rs-fMRI correlation connectivity data
Inputs: rs-fMRI only
Task: Multi-label classification of six AD stages
Dataset: ADRG
Note: Uses functional connectivity from resting-state fMRI

[Enhancing the early detection of Alzheimer's disease using an integrated CNN-LSTM framework: A robust approach for fMRI-based multi-stage classification, Farhan et al. 2025]
Architecture: 18-layer 3D-CNN (spatial) + LSTM (temporal)
Inputs: Raw 4D rs-fMRI time series (no accelerated learning)
Task: Binary, 3-class, and 4-class staging (AD/aMCI/pMCI/HC)
Dataset: ADNI subset (100 subjects)
Performance: "92% Accuracy and AUC across most tasks.
Key Note: This U-Net handles full 4D data without feature learning time, yielding exceptional accuracy

[Deep learning analysis of fMRI data for predicting Alzheimer's Disease: A focus on convolutional neural networks and model interpretability, Zhou et al. 2024]

Clinical

[Neuropsychological decline up to 20 years before incident mild cognitive impairment, Alzheimers Dement., Caselli RJ, Longiusi BT, Burch AC, Chen Y, Su Y, Locke DE, Woodruff BK, Reiman EM. 2024]
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